

watchAny 2.0

Enterprise Architecture, System Specification & Engineering Reference

Release Version:	v2.1.20 (Build 2.1.20+120)
Target Platforms:	Windows Desktop (x64), Android (ARM64-v8a, ARMeabi-v7a, x86_64, Universal APK, AAB)
Primary Tech Stack:	Flutter 3.7.2+, Dart 3.7+, Android Kotlin/Java, JVM ASM Bytecode Patcher
Repository:	Fluffy-The-Penguin/watchAny-2.0
Date of Issue:	July 27, 2026
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Executive Summary:

watchAny 2.0 is a production-grade multi-platform streaming & reading engine designed to dynamically load and execute native R8-compiled Mihon (Tachiyomi) manga extension APKs on Android and patched bytecode JARs on Windows Desktop. Powered by media_kit C++ bindings and an embedded NanoHTTPD REST IPC server, watchAny provides 100% extension compatibility with zero UI stutter.

1. High-Level Architecture & Subsystem Flow

watchAny 2.0 operates as a hybrid application split into a **Flutter Frontend UI Layer** and an **Embedded Native Extension Runtime Engine**. Communication between Dart UI and the underlying Kotlin/Java runtime occurs via a local loopback REST API server operating on port 4567.

Subsystem Layer	Components & Technologies	Responsibility
UI & Presentation Layer	Flutter 3.7+, Riverpod 2.5, WindowManager	Renders fluid 60 FPS UI, manages application state, handles gesture overlays and keyboard navigation.
Video & Reader Engines	MediaKit (libmpv C++), Custom Canvas Virtualizer	Hardware-accelerated video playback and high-throughput virtualized manga page rendering.
Embedded REST IPC Gateway	NanoHTTPD (Port 4567), Kotlin Coroutines	Exposes /api/v1/manga/* REST endpoints to Dart UI, converting async Kotlin execution to JSON payloads.
Android Extension Runtime	ChildFirstPathClassLoader, InjektScope	Loads raw Android extension APKs with parent-first delegation for host singletons and interceptors.
Desktop JVM Runtime	ASM Bytecode Manipulator, DynamicExtensionPatcher	Patches obfuscated extension JARs at runtime, neutralizing R8 throws and injecting lambda initializers.
Network & Interceptor Chain	OkHttp 5.0, Brotli, Cloudflare, UserAgent	Executes remote web requests with complete anti-bot bypass, cookie jars, and rate limiting.

2. Complete Technology Stack & Dependency Catalog

Every third-party package and native library in watchAny 2.0 has been selected to meet explicit performance, compatibility, and stability requirements.

2.1 Frontend & Cross-Platform Dart Packages (pubspec.yaml)

Package	Version	Technical Purpose & System Justification
flutter_riverpod	2.5.1	Compile-time safe state management. Isolates extension catalog state, active player settings, and theme providers.
media_kit	1.2.6	High-level Flutter video player API interfacing with libmpv native C++ core.
media_kit_video	2.0.1	Native Texture widget for video rendering with gesture controls and aspect ratio scaling.
media_kit_libs_video	1.0.7	Bundles pre-compiled native libmpv binaries (.dll, .so) for Windows, Linux, and Android.
drift	2.24.1	Type-safe reactive SQLite database manager for library tracking, history, and user settings.
sqlite3_flutter_libs	0.5.25	Ships updated native SQLite C-libraries to eliminate platform-dependent database discrepancies.
dio & dio_http2_adapter	5.7.0 / 2.1.0	HTTP/2 client for high-speed concurrent page tile fetching and API queries.
flutter_js	0.8.7	Embedded QuickJS C++ JavaScript engine for running sandboxed anime video extractor scripts.
window_manager	0.5.1	Windows native window control API for custom glassmorphism titlebars and borderless frames.
cached_network_image	3.4.1	Disk & memory image caching engine with custom Referer header support for manga page tiles.
workmanager	0.7.0	Android WorkManager wrapper for background library chapter update checks.

2.2 Native Android Dependencies (build.gradle.kts)

Artifact Group & Name	Version	Extension Runtime Functionality
com.squareup.okhttp3:okhttp	5.0.0-alpha.14	Core HTTP network engine required by Mihon extensions for connection pooling & SSL.
com.squareup.okhttp3:okhttp-brotli	5.0.0-alpha.14	Provides native BrotliInterceptor for decompression of modern web responses.
io.reactivex:rxjava	1.3.8	RxJava 1.x runtime for compatibility with legacy Tachiyomi extension source classes.
org.jsoup:jsoup	1.18.3	HTML DOM parsing library used by extensions for scraping manga catalogs.
net.dongliu:apk-parser	2.6.10	Pure-Java APK parser used to extract package info, metadata, and source classes.
org.nanohttpd:nanohttpd	2.3.1	Embedded HTTP server running the REST API bridge inside the Android process.
desugar_jdk_libs	2.1.4	Java 8+ API desugaring for backwards compatibility on older Android devices.

3. Dual-Engine Extension Runtime Architecture

3.1 Android Native ClassLoader Architecture

On Android, extensions are packaged as standard APK files containing compiled DEX bytecode. To ensure singletons and type checks match 100% between the host application and loaded extensions, watchAny implements `ChildFirstPathClassLoader` with **Parent-First Delegation** for host framework packages:

```
// Android Custom ClassLoader Delegation Chain
override fun loadClass(name: String?, resolve: Boolean): Class<*> {
    var c = findLoadedClass(name)

    // 1. System ClassLoader (java.*, android.*)
    if (c == null && systemClassLoader != null) {
        try { c = systemClassLoader.loadClass(name) } catch (_: ClassNotFoundException) {}
    }

    // 2. Force Parent ClassLoader for Host Framework Singletons
    if (c == null && name != null && (
        name.startsWith("eu.kanade.tachiyomi.network.") ||
        name.startsWith("eu.kanade.tachiyomi.source.") ||
        name.startsWith("uy.kohesive.injekt.") ||
        name.startsWith("okhttp3.") ||
        name.startsWith("rx.") ||
        name.startsWith("org.jsoup.")
    )) {
        try { c = parent.loadClass(name) } catch (_: ClassNotFoundException) {}
    }

    // 3. Fallback to Extension APK DEX Search
    if (c == null) {
        c = try { findClass(name) } catch (_: ClassNotFoundException) { super.loadClass(name, resolve) }
    }
    return c
}
```

3.2 Desktop ASM Bytecode Patcher Architecture

On Windows Desktop, Android DEX files cannot run directly on the standard Java Virtual Machine. watchAny utilizes an **ASM Bytecode Manipulation Engine** (`MasterRuntimePatcher.java` & `DynamicExtensionPatcher.java`) to dynamically transform extension classes at load time:

- **ATHROW Neutralization:** Obfuscated R8 validation checks that throw when specific interceptors are missing are replaced with clean signature-matched return opcodes (RETURN for void, ARETURN for objects, IRETURN for integers/booleans).
- **Lambda & Companion Initializer Repair:** R8 optimizes away default zero-argument constructors from companion objects; the patcher dynamically injects missing `<init>()V` constructors.
- **NetworkHelper Method Routing:** Intercepts calls to `getClient()`, `getCloudflareClient()`, and `getDefaultClient()` to inject `DummyInterceptors`.
- **Slug Parser Fallbacks:** Replaces null-memo slug parsing instructions in Asura and Flame Comics extensions with safe URL string concatenation.

4. Local REST API & Network Interceptor Suite

4.1 Embedded NanoHTTPD REST Gateway

The application embeds a lightweight NanoHTTPD REST server running on port 4567. This server exposes unified endpoints for catalog browsing, search queries, manga details, chapter lists, and page image URL extraction:

HTTP Method	Endpoint Route	Description & Payload Behavior
GET	/api/v1/extensions	Returns JSON array of all installed extensions, package names, logos, and source IDs.
GET	/api/v1/manga/popular	Queries popular manga feed for ?sourceId=X&page;=Y.
GET	/api/v1/manga/latest	Queries latest updates feed for ?sourceId=X&page;=Y.
GET	/api/v1/manga/search	Executes catalog search for ?sourceId=X&query;=Q&page;=Y.
GET	/api/v1/manga/details	Fetches full manga metadata (description, author, cover, tags) for ?sourceId=X&mangaUrl;=U.
GET	/api/v1/manga/chapters	Fetches complete chapter list for ?sourceId=X&mangaUrl;=U.
GET	/api/v1/manga/pages	Resolves image URLs for all pages in chapter ?sourceId=X&chapterUrl;=U.

4.2 Complete Network Interceptor Chain

To bypass anti-bot mechanisms and pass extension validation checks, watchAny includes a complete OkHttp interceptor chain:

- UncaughtExceptionHandlerInterceptor:** Catches unexpected network exceptions and wraps non-IOExceptions.
- UserAgentInterceptor:** Automatically injects modern Android/Desktop Chrome User-Agent header strings and Client Hints (Sec-CH-UA).
- CloudflareInterceptor:** Preserves Cloudflare clearance cookies and manages anti-bot challenges.
- RateLimitInterceptor:** Implements per-domain request rate limiting to prevent IP throttling.
- CompressionInterceptor (Brotli, Gzip, Zstd):** Manages response stream decompression algorithms.

5. Build & Release Engineering Pipeline

The release engineering workflow produces **6 distinct artifacts** for distribution across desktop and mobile architectures. Every build follows a strict sequence of version bumping, compilation, packaging, and automated GitHub release deployment.

Target Platform / Architecture	Artifact Type	Output Filename	Approx. Size
Windows Desktop (x64)	Portable Zip Package	watchany_portable_2.1.20.zip	70.7 MB
Android Universal	Universal APK	app-release.apk	105.2 MB
Android ARM 64-bit	Architecture APK	app-arm64-v8a-release.apk	67.9 MB
Android ARM 32-bit	Architecture APK	app-armeabi-v7a-release.apk	40.5 MB
Android x86 64-bit	Architecture APK	app-x86_64-release.apk	42.0 MB
Android Play Store	App Bundle (AAB)	app-release.aab	131.9 MB

5.1 Step-by-Step Release Build Script

```
# Step 1: Update Version Numbers
# - pubspec.yaml: version: 2.1.20+120
# - lib/services/update_service.dart: currentVersion = '2.1.20'
# - installer.iss: AppVersion=2.1.20

# Step 2: Build Windows Desktop Release & Portable Zip
flutter build windows --release
powershell -Command "Compress-Archive -Path 'build\windows\x64\runner\Release\*'
-DestinationPath 'watchany_portable_2.1.20.zip' -Force"

# Step 3: Build Android Universal APK
flutter build apk --release

# Step 4: Build Android Split-per-ABI APKs (ARM64, ARMv7, x86_64)
flutter build apk --release --split-per-abi

# Step 5: Build Android App Bundle (AAB)
flutter build appbundle --release

# Step 6: Commit, Tag & Upload Release via GitHub CLI
git add android/app/src/main/kotlin/ pubspec.yaml lib/services/update_service.dart installer.iss
git commit -m "v2.1.20 - Complete Mihon extension network support with okhttp3.brotli.Brotli
integration"
git push origin main

gh release create v2.1.20 \
watchany_portable_2.1.20.zip \
build/app/outputs/flutter-apk/app-release.apk \
build/app/outputs/flutter-apk/app-arm64-v8a-release.apk \
build/app/outputs/flutter-apk/app-armeabi-v7a-release.apk \
build/app/outputs/flutter-apk/app-x86_64-release.apk \
build/app/outputs/bundle/release/app-release.aab \
--title "v2.1.20" \
--notes "v2.1.20 - Complete Mihon extension network support with okhttp3.brotli.Brotli
integration"
```

6. Diagnostics & Historical Bug Resolution Reference

Exception / Failure Symptom	Root Cause Analysis	Permanent Resolution Applied
UncaughtExceptionHandler must be present in default client	Validation check inside R8 extension bytecode failed due to missing interceptor or duplicate ClassLoader instance.	Added missing interceptors to NetworkHelper and enforced Parent-First delegation in ChildFirstPathClassLoader.kt.
NoClassDefFoundError: Lokhttp3/CompressionInterceptor;	okhttp3.CompressionInterceptor class was missing from Android application package.	Created CompressionInterceptor.kt in package okhttp3 in host app.
NoClassDefFoundError: Lokhttp3/brotli/Brotli;	okhttp3.brotli.Brotli object was missing from Android application package.	Created Brotli.kt in package okhttp3.brotli delegating to native BrotliInterceptor.
Type okhttp3.brotli.BrotliInterceptor is defined multiple times	Duplicate class definition between custom source and com.squareup.okhttp3:okhttp-brotli library.	Removed duplicate BrotliInterceptor.kt file and referenced library's native BrotliInterceptor.
VerifyError on neutralized ATHROW	ASM patcher emitted ARETURN inside void (V) return methods on Desktop.	Updated DynamicExtensionPatcher.java to emit signature-matched return instructions.